PROJECT REPORT

FOR

Foundations of Artificial Intelligence

AI-Based Personalized Financial Planning System

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report "**AI-Based Personalized Financial Planning System**" is the bonafide work of " Nagulan A ([240701344]) & Mohan Prasath P ([240701328]) " who carried out the project work under my supervision.

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ABSTRACT:

The **AI-Based Personalized Financial Planning System** is an intelligent, web-based financial decision-support application designed to help individuals manage their finances and make informed financial decisions. The system integrates classical Artificial Intelligence techniques, including Expert Systems, Fuzzy Logic, Financial Health Scoring, and Genetic Algorithms, to provide personalized financial recommendations based on individual financial conditions, goals, and risk tolerance.

The system collects and analyzes users' financial information, including income, expenses, debts, assets, savings, and investment goals. An Expert System evaluates the user's financial condition using predefined rules, while Fuzzy Logic is used to assess risk tolerance and handle uncertainty in financial decision-making. The Financial Health Scoring module evaluates the user's overall financial well-being, and the investment capacity analysis determines the amount available for savings and investments. A Genetic Algorithm is employed to optimize portfolio allocation based on financial constraints and the user's risk profile.

The system is developed using **React, Vite, and Tailwind CSS** for the frontend, **FastAPI and Python** for the backend, and **MongoDB** for database management. It provides features such as financial health assessment, goal tracking, personalized investment recommendations, portfolio optimization, scenario simulation, and financial report generation with PDF export.

The proposed system aims to provide a transparent, explainable, and user-friendly approach to personal financial planning by integrating multiple classical AI techniques into a unified platform. It helps users understand their financial position, evaluate investment options, and plan their financial goals. The system is intended for educational and decision-support purposes and does not guarantee investment returns or replace professional financial advice.

1. INTRODUCTION

Personal financial planning is an essential process that helps individuals manage their income, expenses, savings, debts, and investments to achieve their financial goals. Effective financial planning requires individuals to understand their current financial condition, evaluate their spending habits, maintain emergency savings, and make appropriate investment decisions. However, managing all these financial aspects simultaneously can be difficult without proper guidance and analytical tools.

The **AI-Based Personalized Financial Planning System** is an intelligent, web-based application designed to assist individuals in managing their finances through Artificial Intelligence techniques. The system integrates financial assessment, risk profiling, investment planning, and portfolio optimization into a unified platform. It analyzes user-provided financial information, including income, expenses, debts, assets, savings, and financial goals, to generate personalized recommendations.

The proposed system uses classical Artificial Intelligence techniques such as **Expert Systems, Fuzzy Logic, Financial Health Scoring, and Genetic Algorithms** to support financial decision-making. The Expert System evaluates the user's financial condition using predefined rules and financial parameters. Fuzzy Logic is used to assess risk tolerance and handle uncertainty in financial preferences. The Financial Health Scoring module evaluates the overall financial condition of the user, while investment capacity analysis determines the amount available for savings and investments.

The system also incorporates a Genetic Algorithm to optimize investment portfolio allocation according to the user's financial constraints, investment goals, and risk profile. It evaluates different asset allocation combinations and identifies suitable portfolio distributions based on the configured constraints. This approach helps users understand how their available funds can be distributed among different investment categories, such as equity, bonds, gold, and cash.

The proposed application provides several features, including financial profile management, goal tracking, financial health assessment, risk profiling, personalized investment recommendations, portfolio optimization, and financial report generation. It also supports scenario simulation, allowing users to compare different financial decisions and understand their potential effects. The system generates explainable reports containing financial summaries, recommendations, and actionable insights, which can be exported in PDF format.

The application is developed using **React, Vite, and Tailwind CSS** for the frontend, **FastAPI and Python** for the backend, and **MongoDB** for database management. Libraries such as NumPy, SciPy, scikit-fuzzy, and DEAP are used to implement the financial analysis and AI-based optimization techniques. The system follows a modular architecture in which the user interface, backend services, database, and financial intelligence components work together to process financial information and generate recommendations.

The primary aim of this project is to provide a transparent, explainable, and personalized approach to financial planning. By integrating multiple classical AI techniques, the system helps users understand their financial health, evaluate their investment capacity, manage financial goals, and explore suitable investment strategies. The recommendations are based on user-provided information and configured assumptions. The system is intended for educational and financial decision-support purposes and does not guarantee investment returns or replace professional financial advice.



Fig 1.1 Forward Chaining Algorithm

2. LITERATURE REVIEW

Traditional Personal Financial Management Systems:

Traditional financial management applications, such as **Mint and YNAB**, focus on expense tracking, budgeting, and financial reporting. These applications help users monitor their spending habits and manage their budgets. However, they provide limited personalized investment recommendations and forward-looking financial guidance.

Rule-Based Expert Systems in Financial Planning:

Rule-based Expert Systems use predefined **IF-THEN rules** to evaluate financial conditions and support decision-making. These systems can analyze income, expenses, savings, and debt to generate financial recommendations. Their transparent reasoning makes them suitable for explainable financial planning.

Fuzzy Logic and Financial Risk Assessment:

Fuzzy Logic is used to handle uncertainty in financial decision-making. It helps assess an individual's risk tolerance by considering different financial factors. This approach supports the classification of users into conservative, moderate, and aggressive risk categories, enabling personalized investment recommendations.

Genetic Algorithms for Portfolio Optimization:

Genetic Algorithms are optimization techniques inspired by natural selection. They are used in portfolio optimization to identify suitable asset allocations based on financial constraints and risk preferences. Through selection, crossover, and mutation, the algorithm searches for suitable investment combinations.

Explainable AI in Financial Decision Support:

Explainable Artificial Intelligence focuses on making AI-generated recommendations understandable to users. Rule-based reasoning and transparent optimization processes help

users understand the factors influencing financial decisions. This approach is important in financial planning, where users need clear explanations of recommendations.

Research Gap:

Existing financial applications primarily focus on tracking and reporting rather than personalized financial advice. Financial profiling, risk assessment, and portfolio optimization are often handled separately. The proposed **AI-Based Personalized Financial Planning System** addresses these limitations by integrating Expert Systems, Fuzzy Logic, Financial Health Scoring, and Genetic Algorithms into a unified and explainable platform.

3. PROPOSED SYSTEM

The proposed **AI-Based Personalized Financial Planning System** is a web-based application designed to provide intelligent, personalized, and explainable financial recommendations. The system integrates Expert Systems, Fuzzy Logic, Financial Health Scoring, and Genetic Algorithms to analyze users' financial conditions and support informed financial decision-making.

The system collects financial information, including income, expenses, debts, assets, savings, and financial goals. It evaluates the user's financial health, assesses risk tolerance, estimates investment capacity, and generates personalized investment recommendations.

The Expert System evaluates financial conditions using predefined rules, while Fuzzy Logic is used for risk profiling. The Financial Health Scoring module calculates an overall financial health score, and the investment capacity module estimates the amount available for investment. The Genetic Algorithm optimizes portfolio allocation based on the user's risk profile and financial constraints.

The system also provides features such as financial profile management, goal tracking, scenario simulation, portfolio rebalancing, and financial report generation. It generates explainable recommendations to help users understand the reasoning behind financial decisions. Reports can also be exported in PDF format for future reference.

The application is developed using React, Vite, and Tailwind CSS for the frontend, FastAPI and Python for the backend, and MongoDB for database management. Libraries such as NumPy, SciPy, scikit-fuzzy, and DEAP are used to implement the AI-based financial analysis and optimization modules.

Major Components

1. **Financial Profile Management**
    - Collects and manages users' financial information.
    - Tracks income, expenses, debts, and assets.
    - Supports financial goal creation and progress monitoring.
2. **Expert System**
    - Evaluates financial conditions using predefined rules.
    - Identifies financial issues and provides recommendations.
    - Supports rule-based financial decision-making.
3. **Fuzzy Logic Module**
    - Evaluates users' financial risk tolerance.
    - Handles uncertainty in financial preferences.
    - Classifies users into different risk categories.
4. **Financial Health Scoring**
    - Evaluates the user's overall financial condition.
    - Calculates a financial health score.
5. **Investment Capacity Analysis**
    - Estimates the amount available for investment.
    - Considers income, expenses, debt, and emergency fund requirements.
    - Helps determine suitable investment amounts.
6. **Genetic Algorithm**
    - Optimizes investment portfolio allocation.
    - Considers financial constraints and risk profiles.
    - Evaluates different asset allocation combinations.
7. **Recommendation and Reporting Module**
    - Generates personalized financial recommendations.
    - Generates financial reports with explanations and PDF export.

Working Principle

The system follows a sequential process to generate personalized financial recommendations:

1. **Financial Data Collection:** Collects the user's financial information, including income, expenses, debts, assets, and financial goals.
2. **Financial Health Assessment:** Evaluates the user's financial condition and calculates a financial health score.
3. **Emergency Fund Evaluation:** Identifies emergency fund requirements and evaluates the user's financial preparedness.
4. **Investment Capacity Estimation:** Determines the amount available for investment based on the user's financial condition.
5. **Risk Assessment:** Uses Fuzzy Logic to evaluate the user's risk tolerance.
6. **Financial Constraint Application:** Applies financial constraints to ensure that the portfolio allocation considers the user's financial capacity and risk profile.
7. **Portfolio Optimization:** Uses a Genetic Algorithm to identify suitable investment allocations.
8. **Portfolio Ranking:** Ranks the generated portfolios based on their suitability and configured utility criteria.
9. **Recommendation Generation:** Generates personalized investment recommendations and explainable financial reports.

3.1 ARCHITECTURE DIAGRAM

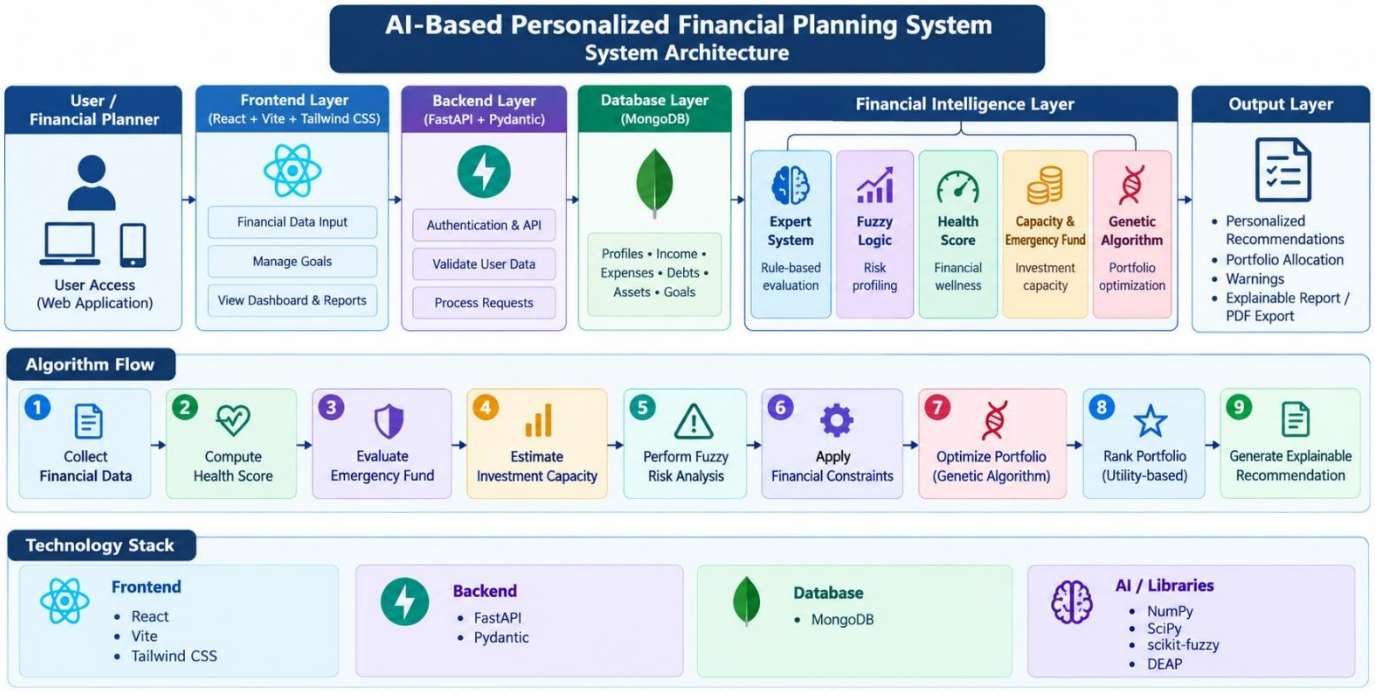


Figure 3.1: System Architecture of AI-Based Personalized Financial Planning System

4. MODULES DESCRIPTION

The AI-Based Personalized Financial Planning System is organized into two major modules that work together to provide an intelligent, personalized, and explainable financial planning environment. The first module manages user interaction, financial data input, goal tracking, and result visualization. The second module handles financial analysis, AI-based reasoning, risk assessment, and portfolio optimization. The integration of these modules enables users to evaluate their financial conditions and receive personalized investment recommendations.

4.1 MODULE 1 – USER INTERFACE MODULE

The User Interface Module forms the primary interaction layer of the system. It provides a responsive web application through which users can enter financial information, manage financial goals, view financial health scores, and access personalized investment recommendations.

The interface is developed using React, Vite, and Tailwind CSS. It communicates with the backend through FastAPI to submit financial data and retrieve the results generated by the financial intelligence modules.

The interface collects information such as income, expenses, debts, assets, savings, and investment goals. It displays the results of financial analysis through dashboards, charts, and reports. Users can also monitor their financial goals, view optimized portfolio allocations, and export financial reports in PDF format.

The user interface is kept separate from the financial reasoning and optimization logic. This separation allows the interface to be modified or enhanced without affecting the underlying financial intelligence modules.

Main functions:

- User interaction and dashboard management.
- Financial data input and profile management.
- Financial goal creation and tracking.
- Display of financial health scores and risk profiles.
- Financial report generation and PDF export.

4.2 MODULE 2 – INFERENCE ENGINE AND KNOWLEDGE BASE MODULE

The Financial Intelligence and Optimization Module is responsible for analyzing users' financial information and generating personalized financial recommendations. It is implemented using Python and FastAPI, with MongoDB used for storing financial profiles, income, expenses, debts, assets, and financial goals.

The module integrates Expert Systems, Fuzzy Logic, Financial Health Scoring, investment capacity analysis, and Genetic Algorithms to perform financial assessment and portfolio optimization.

The Expert System evaluates financial conditions using predefined rules, while Fuzzy Logic assesses the user's risk tolerance. The Financial Health Scoring component calculates the user's overall financial health score, and the investment capacity component estimates the amount available for investment after considering financial obligations and emergency fund requirements.

The Genetic Algorithm, implemented using the DEAP library, optimizes portfolio allocation according to the user's risk profile and financial constraints. The module evaluates different portfolio combinations and ranks them using utility-based criteria. Finally, it generates personalized recommendations and explainable financial insights, which are returned to the user interface.

Main functions:

- Financial data validation and processing.
- Rule-based financial evaluation using Expert Systems.
- Risk profiling using Fuzzy Logic.
- **Portfolio optimization using Genetic Algorithms.**
- Derivation of new facts
- Fired-rule tracking.
- Scenario simulation and financial analysis.

5. IMPLEMENTATION AND RESULTS

The AI-Based Personalized Financial Planning System is developed as a web-based financial decision-support application that integrates a responsive user interface with AI-based financial analysis and portfolio optimization. The system is developed using React, Vite, and Tailwind CSS for the frontend, FastAPI and Python for the backend, and MongoDB for database management.

The application consists of separate components for user authentication, financial profile management, goal tracking, financial assessment, risk profiling, investment recommendations, and portfolio optimization. The financial intelligence layer integrates Expert Systems, Fuzzy Logic, Financial Health Scoring, and Genetic Algorithms to analyze user data and generate personalized recommendations.

When the user submits financial information, the backend processes and validates the data. The financial intelligence modules evaluate the user's financial health, assess risk tolerance, calculate investment capacity, and optimize portfolio allocation using a Genetic Algorithm. The resulting recommendations and financial insights are displayed through the user interface.

The system also provides a financial health dashboard, portfolio allocation visualization, scenario simulation, monthly financial reports, and PDF export functionality. These features help users understand their financial condition and evaluate different financial planning decisions.

5.1 EXPERIMENTAL SETUP

The system is developed using Python 3.10+ for backend processing and AI-based financial analysis. The frontend is developed using React.js, Vite, and Tailwind CSS to provide a responsive user interface. FastAPI is used to develop backend APIs, while MongoDB is used to store financial profiles, income, expenses, debts, assets goals.

The financial intelligence layer uses NumPy and SciPy for numerical calculations, scikit-fuzzy for fuzzy logic-based risk assessment, and DEAP for Genetic Algorithm-based portfolio optimization.

The Expert System evaluates financial conditions using predefined rules, while the Financial Health Scoring module assesses the user's financial wellness. The investment capacity module estimates the available investment amount after considering financial obligations and emergency fund requirements.

The Genetic Algorithm generates and evaluates different portfolio allocations based on financial constraints and risk profiles. The resulting portfolios are ranked using utility-based criteria to support personalized investment recommendations.

The experimental setup includes scenarios such as:

- Evaluates user financial information and calculates a financial health score.
- Uses Fuzzy Logic to assess risk tolerance and classify users into different risk categories.
- Allows users to compare different financial decisions and examine their potential effects.
- Generates personalized financial reports containing financial insights, recommendations, and portfolio information.

4.1 RESULTS

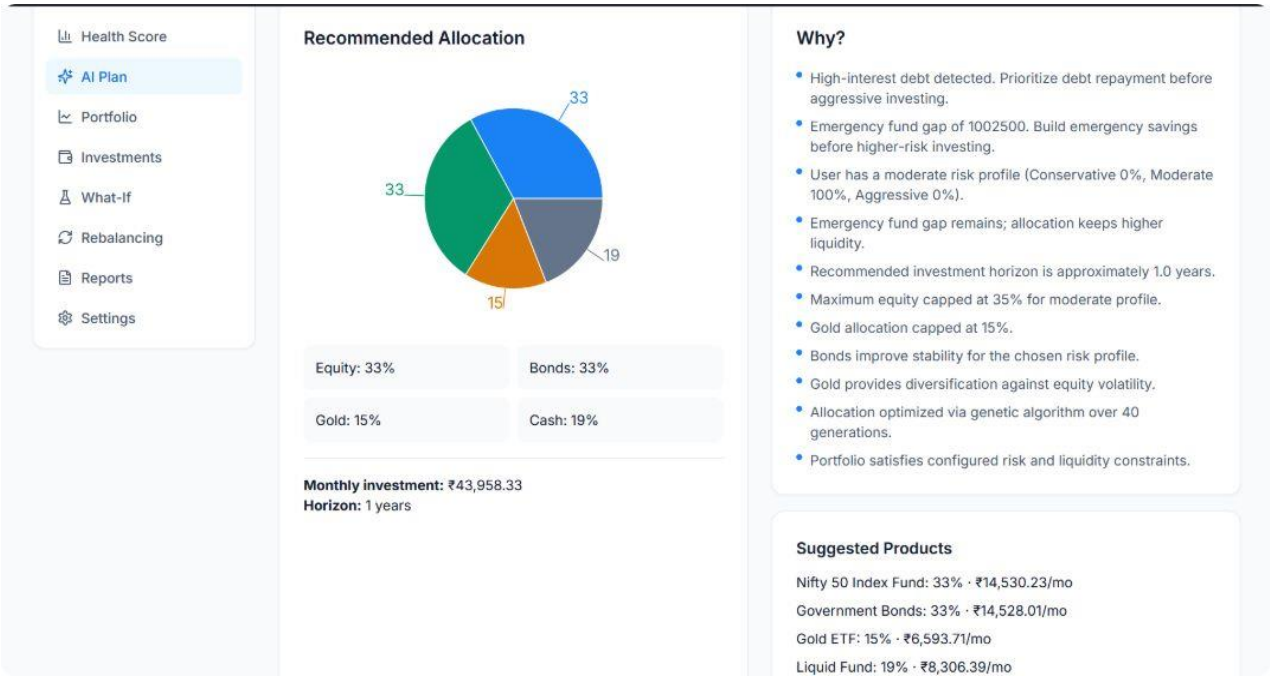


Figure 5.2.1 Main Application Window

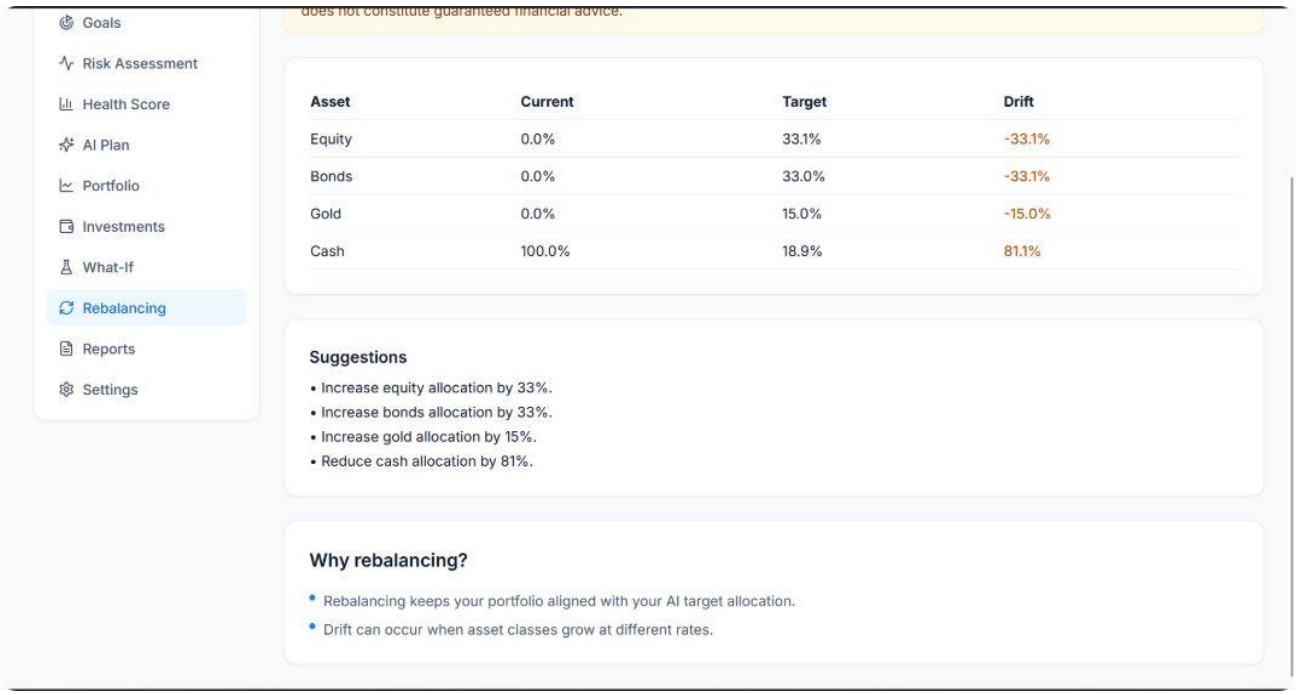


Figure 5.2.2 Portfolio Allocation & Recommendations

5. CONCLUSION AND FUTURE WORK

The **AI-Based Personalized Financial Planning System** is an intelligent and explainable web-based application developed to support personalized financial planning and investment decision-making. The system integrates **React, FastAPI, Python, MongoDB, Expert Systems, Fuzzy Logic, Financial Health Scoring, and Genetic Algorithms** to analyze users' financial conditions and generate personalized recommendations.

The system combines financial profile management, goal tracking, financial health assessment, risk profiling, investment capacity analysis, and portfolio optimization into a unified platform. The integration of classical AI techniques helps users evaluate their financial conditions, understand their risk tolerance, and explore suitable investment strategies. The use of explainable reasoning allows users to understand the factors influencing the recommendations.

FUTURE WORK

In the future, the system can be enhanced by integrating real-time financial market data to provide updated investment analysis and portfolio recommendations. The financial knowledge base can also be expanded with additional rules to cover a wider range of financial situations and investment strategies.

Advanced AI techniques can be explored to improve financial forecasting, risk assessment, and personalized recommendations. The system can also be extended with additional investment options, improved scenario simulation, and automated portfolio rebalancing based on changing financial goals and market conditions.

Further enhancements may include integration with banking APIs for automated financial data collection, mobile application support, and improved visualization of financial trends. These enhancements can improve the system's functionality, scalability, and usefulness while maintaining transparency and explainability in financial decision-making.

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